Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Computer Science and Engineering (Data Science)

Experiment 7

(ETL)

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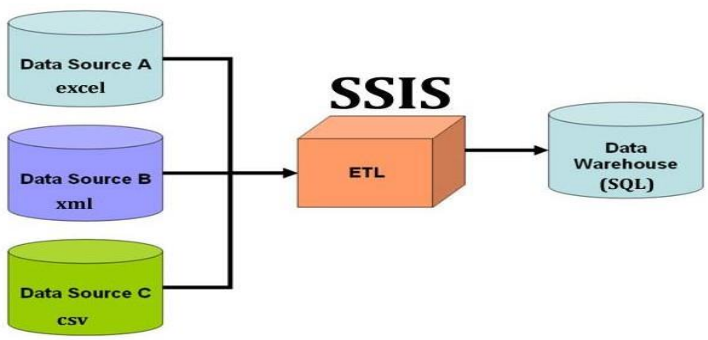
Aim: - Perform ETL using SSIS toll and SQL server.

Theory: -

SQL Server Integration Service (SSIS) is a component of the Microsoft SQL Server database software that can be used to execute a wide range of data migration tasks. SSIS is a fast & flexible data warehousing tool used for data extraction, loading and transformation like cleaning, aggregating, merging data, etc.
It makes it easy to move data from one database to another database. SSIS can extract data from a wide variety of sources like SQL Server databases, Excel files, Oracle and DB2 databases, etc.



What is an ETL process?

ETL stands for Extraction, Transformation and Loading. It is a process in data warehousing to extract data, transform data and load data to final source. ETL covers a process of how the data are loaded from the source system to the data warehouse. Let us briefly describe each step of the ETL process.

Extraction

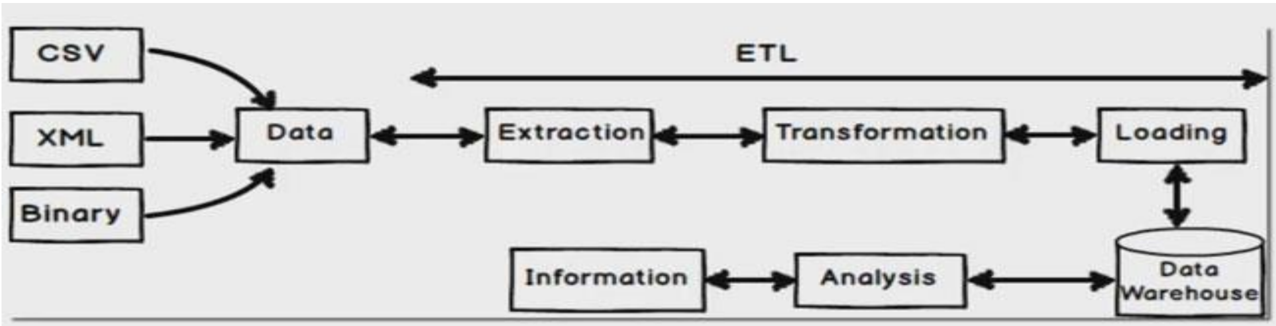
Extraction is the first step of ETL process where data from different sources like txt file, XML file, Excel file or various sources collected.

Transformation

Transformation is the second step of ETL process where all collected data is being transformed into same format i.e. format can be anything as per our requirement before loading it to data-warehouse i.e. it may be data-type format, data merge format, splitting format, alphabet joining format, currency format etc.

Loading

Final step of ETL process, the big chunks of data which is collected from various sources and transformed then finally load to our data warehouse.



ETL process with SSIS Step by Step using example

We do this example by keeping baskin robbins (India) company in mind i.e. customer data which is maintained by small outlet in an excel file and finally sending that excel file to USA (main branch) as total sales per month. This data is necessary at headquarters (main branch) to track performance of each outlet. So here also we will do same thing i.e. We will collect customer product purchase sales data from small-small outlet (In an Excel Format) - Extraction

Since baskin robbins is located in USA we need to convert or transform product purchase amount to USD currency and we will also convert product name to uppercase for unique representation - Transformation

Finally loading this transformed data to database / data warehouse (SQL Server Database) - Loading

Step 1 :- Extract Data From Excel File

Before you read this steps kindly make sure you have installed Microsoft business intelligence along with SQL Server.

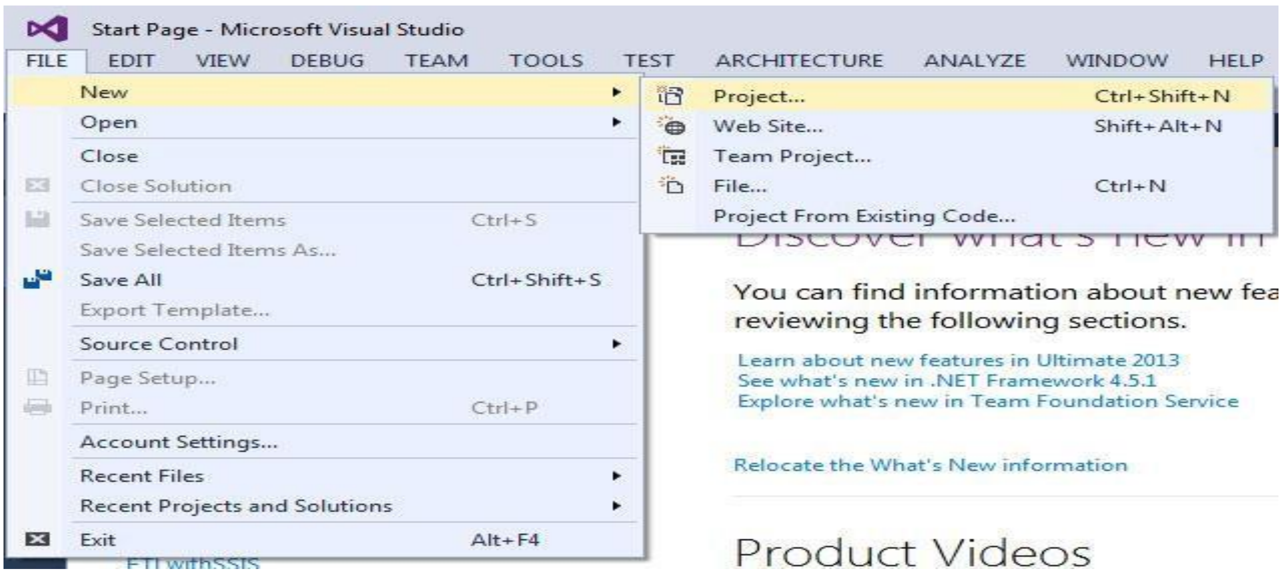
I thin step we will create a simple excel file with columns names as CustomerCode, Customer Name, Product Purchase, Quantity, Amount, CustomerVisitedDate respectively.

Add some data as shown in below image.

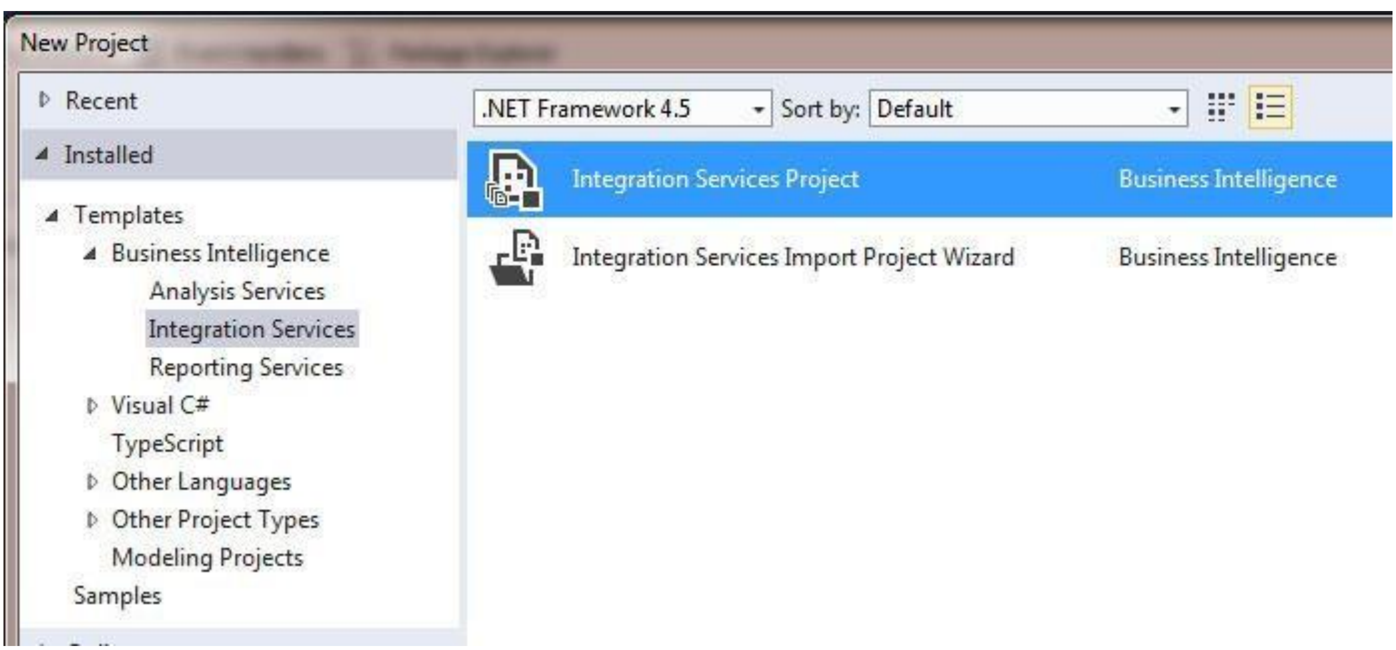
CustomerCode	CustomerName	ProductPurchase	Quantity	Amount	CustomerVisitedDate
101	Shiv	Café ice cream	2	200	1/6/2016
102	Shaam	caramel icecream	2	250	2/6/2016
103	Allwyn	violette icecream	8	1523	1/6/2016
104	Khadak	cassis icecream	3	750	1/6/2016
105	Robin	chocolate icecrea	5	1000	2/6/2016
106	Ajay	pista icecream	4	800	1/6/2016
107	Sukesh	mango icecream	1	150	1/6/2016
108	Praful	Café ice cream	9	2250	2/6/2016

Give a name and save it your computer.

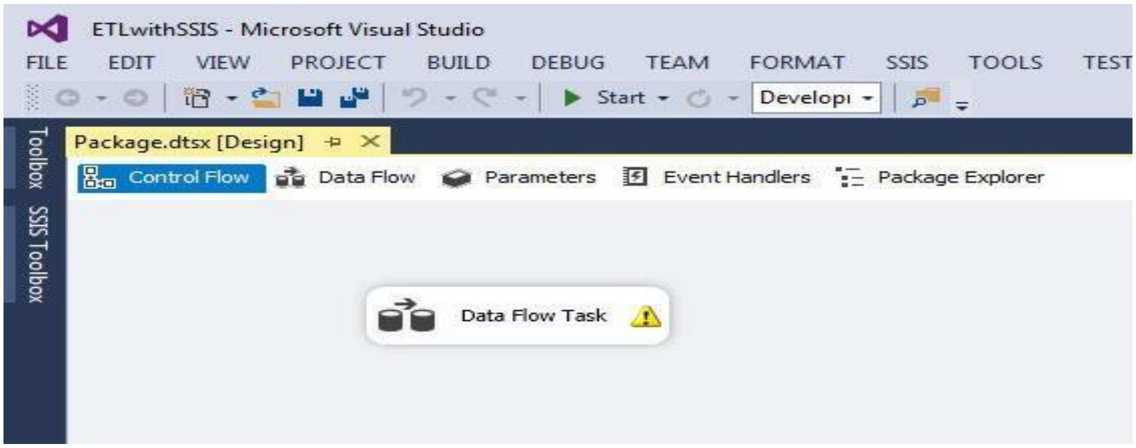
Now open your SQL Server data tools if you don't have SQL server data tools installed i request to read our first article i.e <u>How to install MSBI step by step</u>. if you already MSBI data tool just open it and go to FILE -> NEW -> PROJECT.



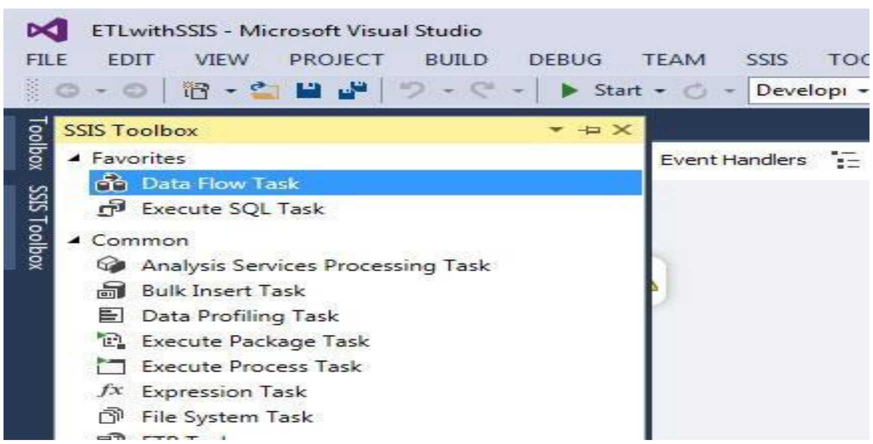
Since we are doing ETL process and that process comes under SSIS so we need to create Integration Services so choose that -> Integration Service Project.



Give a name and create a project.



Now go to SSIS Toolbox and drag and drop "Data Flow Task" to control panel as show in below image.

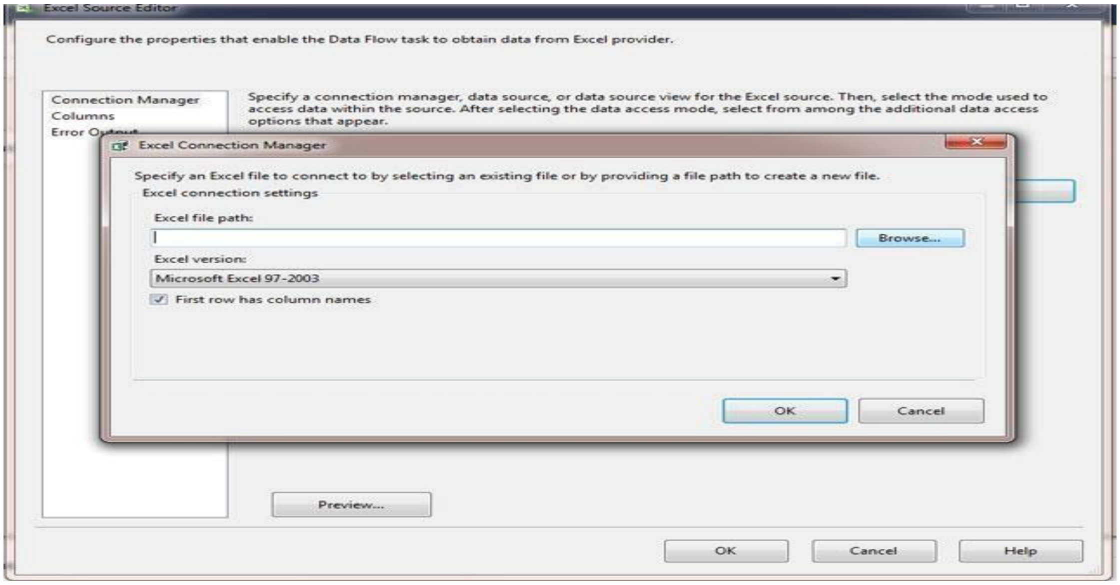


Just double click on "Data Flow Task" to take you to "Data Flow".

Now go to SSIS Toolbox and from Other Sources tab just drag and drop Excel sources. Why excel source because our initial data which we want to extract it is in excel format.

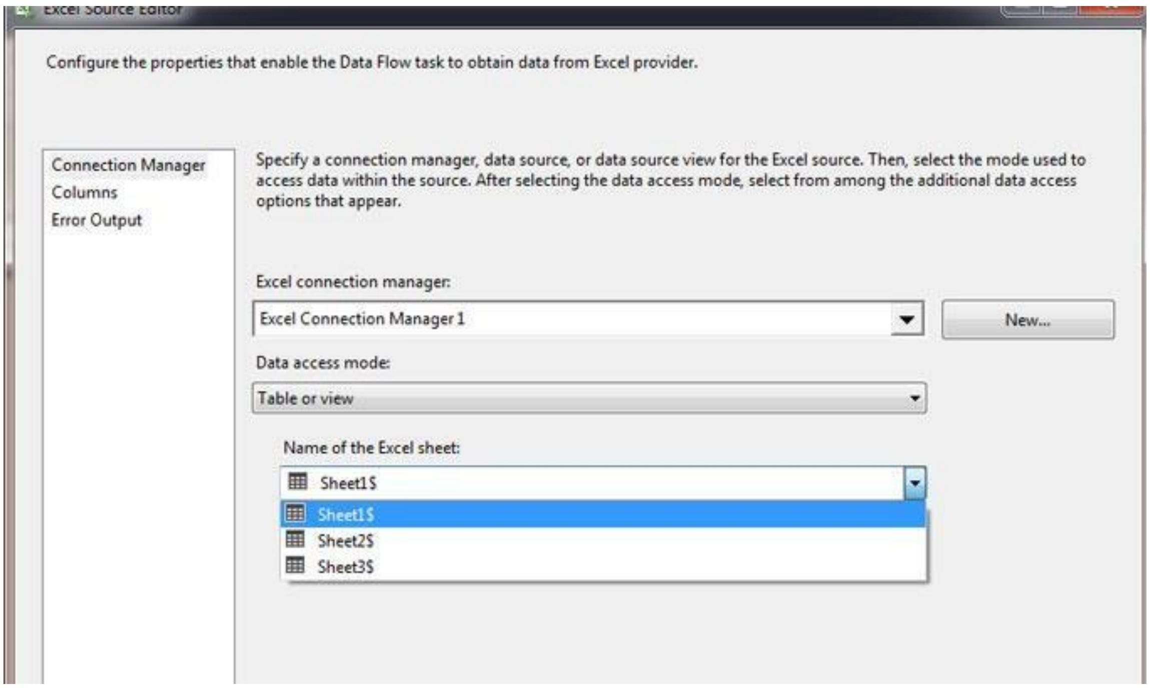
So just drag excel file as shown below image and right click and rename it so that if any developer reads it can easily able to understand.

Now right click on that excel source -> Edit -> Click on New button - Browse Excel file from your computer as shown in below image.



Since our first column of excel file is having column names so we need to check this below check box as you see in above image.

Now select Data access mode as "Table on view" then select Excel sheet name from drop down.



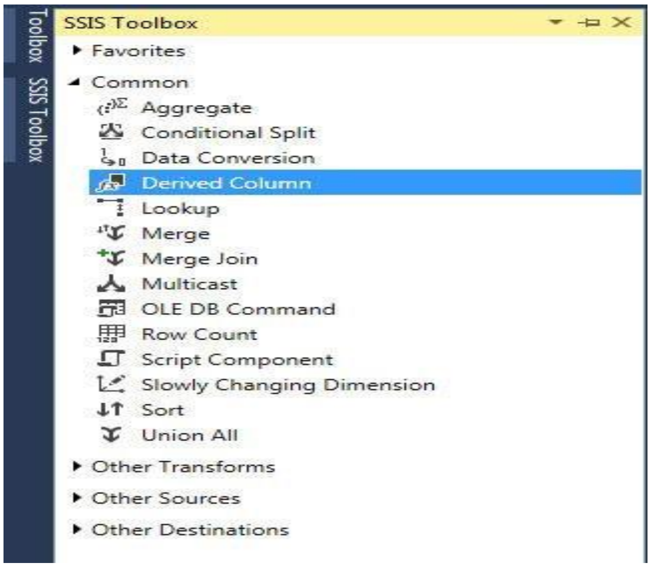
If you want to preview you can also do that by clicking preview button.

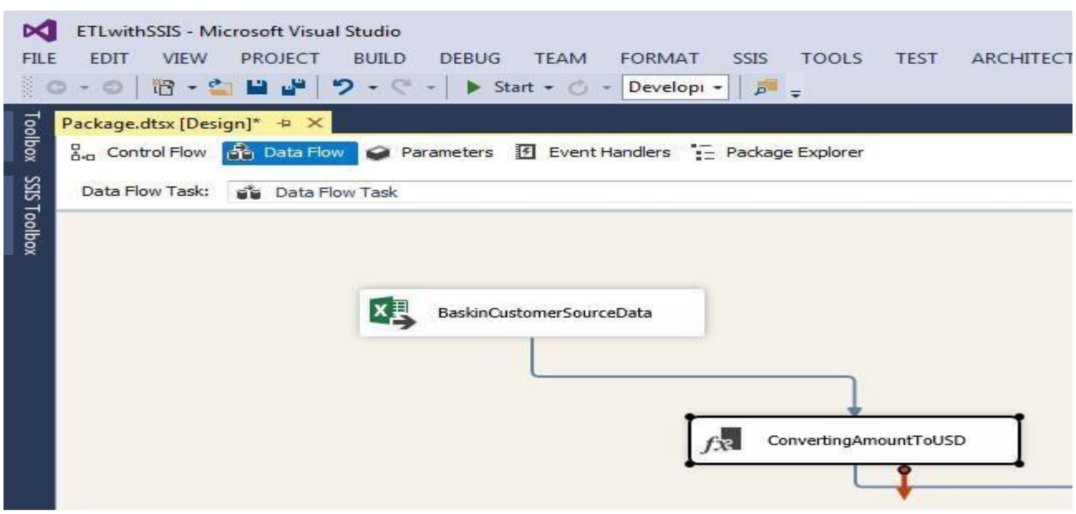
Finally click on OK button. So now yours excel source is ready. It means we have successfully extracted in excel data file to SSIS excel data source.

Step 2: Transform Data (Convert to US currency and Upper case)

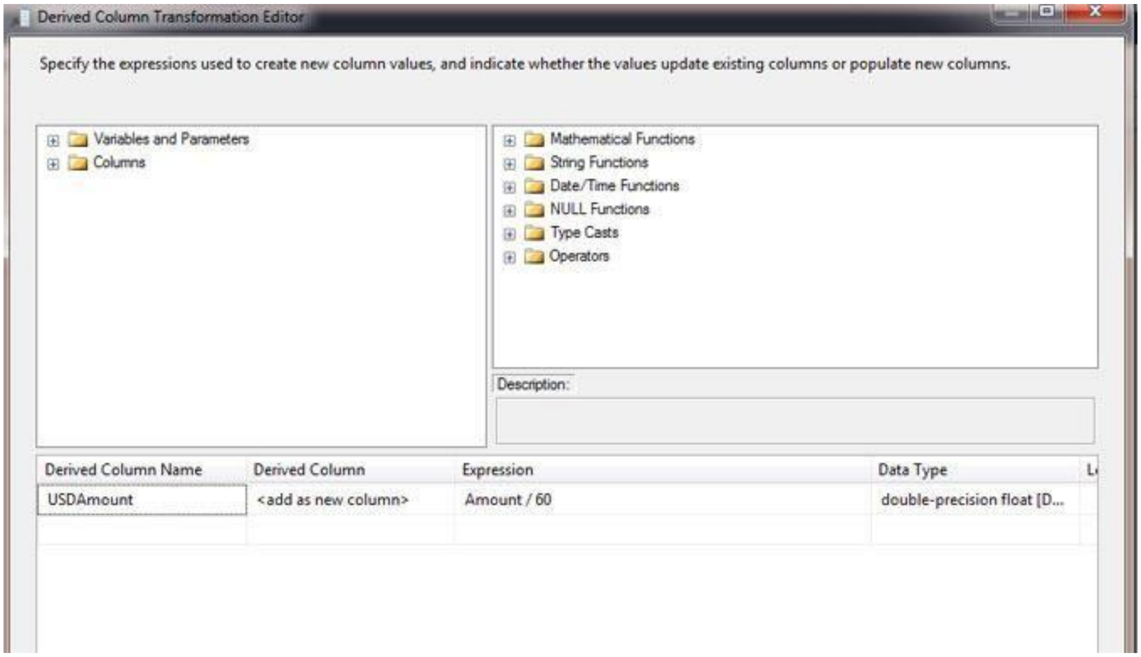
As you now in excel file we have column name called "Amount" and that amount is in Indian currency. So we need to convert that Indian amount to USD amount so to do that we will drag and drop "Derived Column" from SSIS toolbox.

Now if you see on Excel Source file box there are two arrows "Red" and Blue". Just drag that "Blue" arrow and join it to "Derived column" as shown in below image and rename that "Derived column".





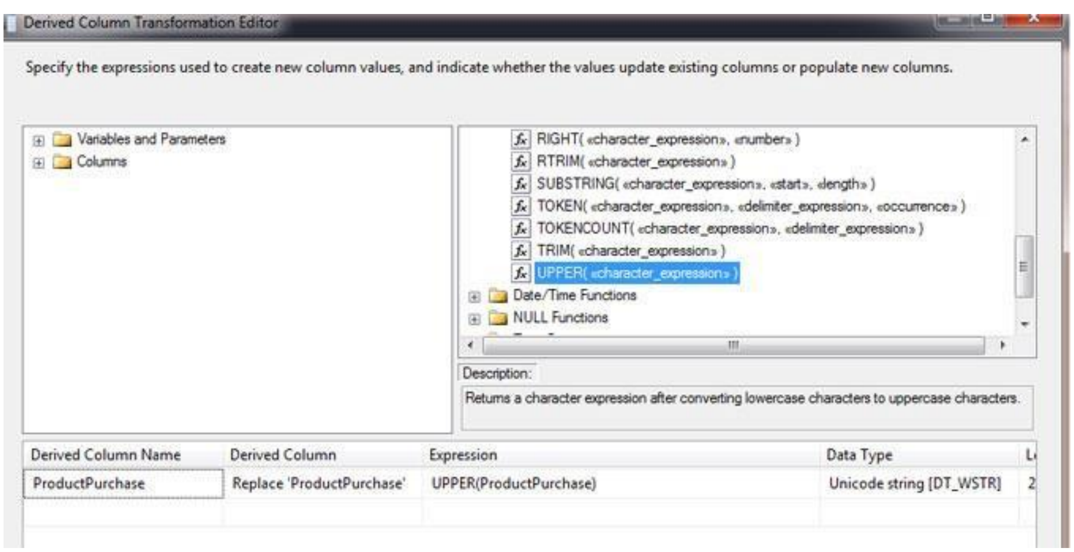
Now right click on "Derived column" i.e ConvertingAmounttoUSD and click on Edit as show below image.



Now in the below Derived Column Name give a new column name "USDAmount" and Derived column select "add a new column" and in the Expressing give the Formula i.e. "Excel Column (Amount / 60)" and Data-type select as double precision float.

Why "Amount / 60" this expression is it because we want to current Indian Amount to USD currency so that's why we an added a new column "USDAmount" and added calculated values to this new column. Finally save it.

We have now USD amount, next step we need stand representation of product name i.e. in UPPER CASE so for that we will add another "Derived Column" same way. Now here we need to drag "Blue" arrow from "ConvertingAmounttoUSD" to new derived column ("CapitalProductName") as shown below image.



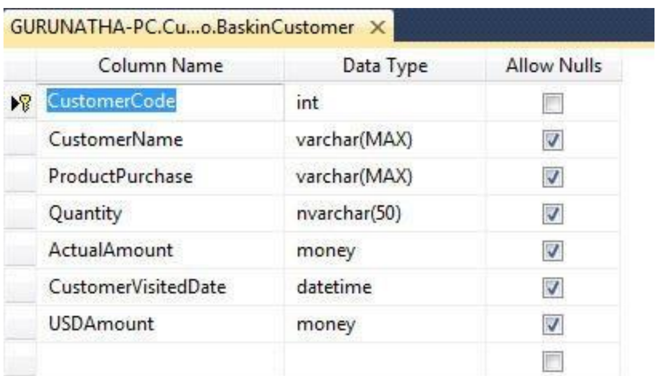
Same way right clicks on "CapitalProductName" go to EDIT choose column "ProductPurchase" from columns section and drag it to "Derive Column Name. In the Derived Column select "Replace 'ProductPurchase'.

Expression you can either choose from string function of you can type it i.e. UPPER (Column name). finally click on OK button.

So as you can see we have completed data transformation part i.e. Convert Amount to USD and Changed Product Names to UPPER case.

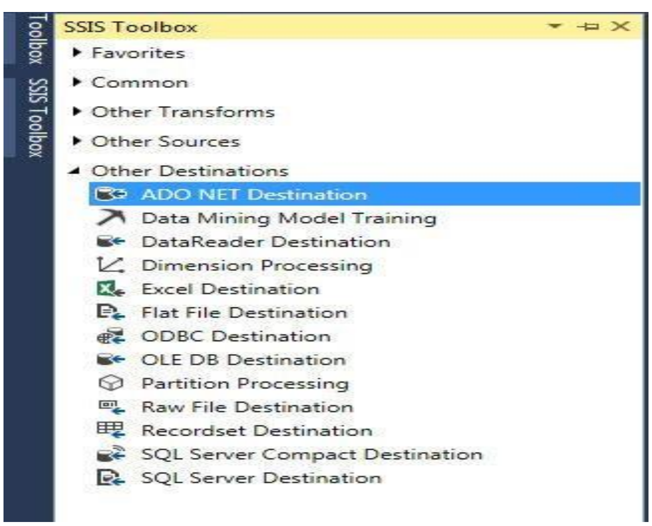
Step 3: Loading Data to SQL Server

Before you start this step just open up your SQL server management studio and create a new database if need or just a new table with same excel column names as shown below image

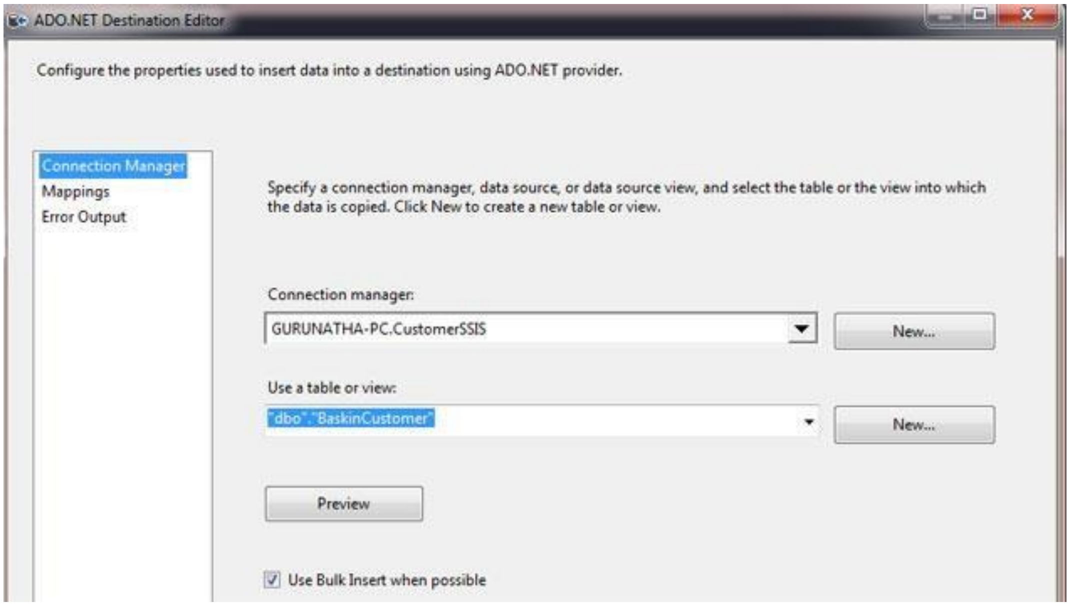

GURUNATHA-PC.Cu...o.BaskinCustomer

Column Name	Data Type	Allow Nulls
CustomerCode	int	☐
CustomerName	varchar(MAX)	☑
ProductPurchase	varchar(MAX)	☑
Quantity	nvarchar(50)	☑
ActualAmount	money	☑
CustomerVisitedDate	datetime	☑
USDAmount	money	☑
		☐

Now go to your SSDT and from SSIS toolbox under Other Destination select "ADO.NET Destination “and drag it to "Data Flow" and drag arrow from "CapitalProductName" to "ADO.NET Destination".



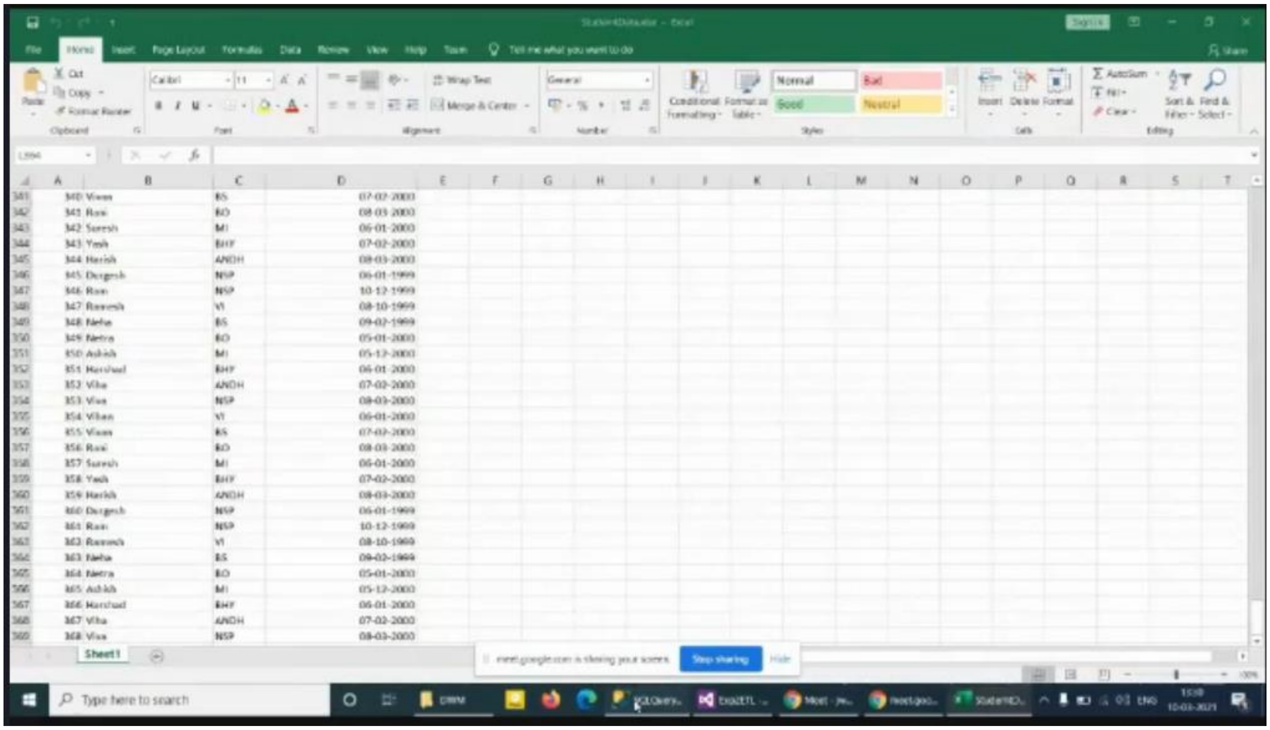
Right click and click on edit add your server name if you do not know your extract server name go to sql server management studio -> File -> click on connect and copy that server name. Now come back to SSDT and click New button -> Again New -> give here server name. Once you give server name automatically database dropdown will populate. Select database and click on OK button. Finally choose table as shown in below image and save it (OK button).



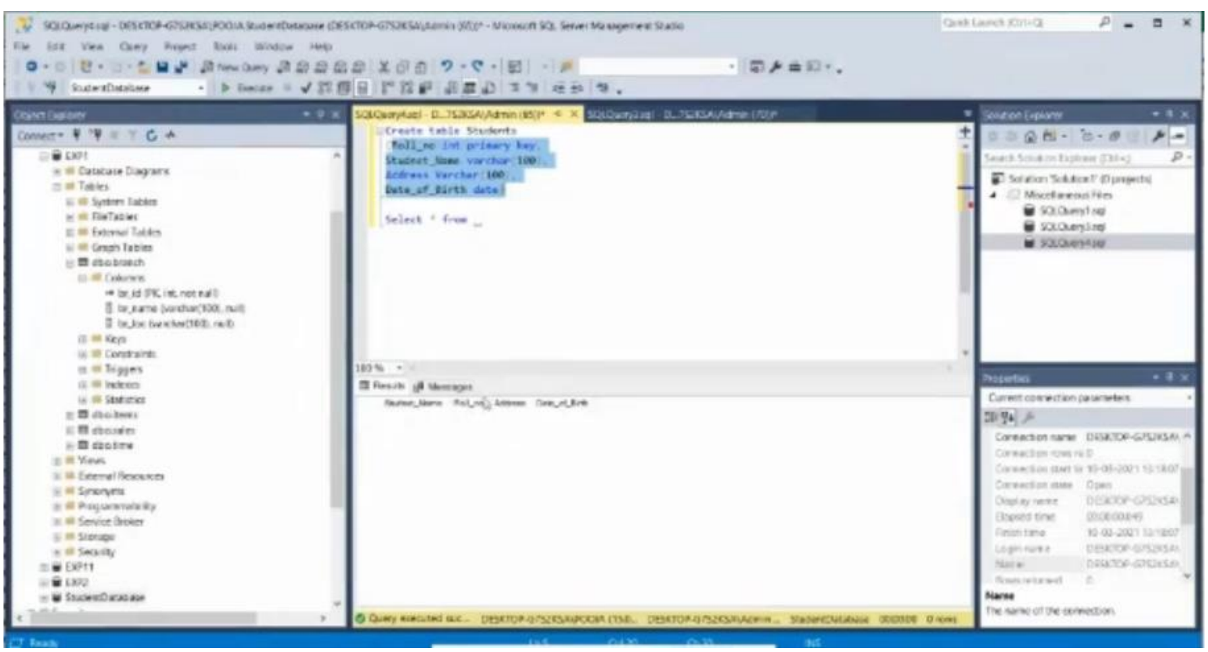
Lab Assignments to complete in this session': -

1. Download a dataset for performing ETL operations
2. Perform Extraction, Transformation and Loading Operations on the above data set

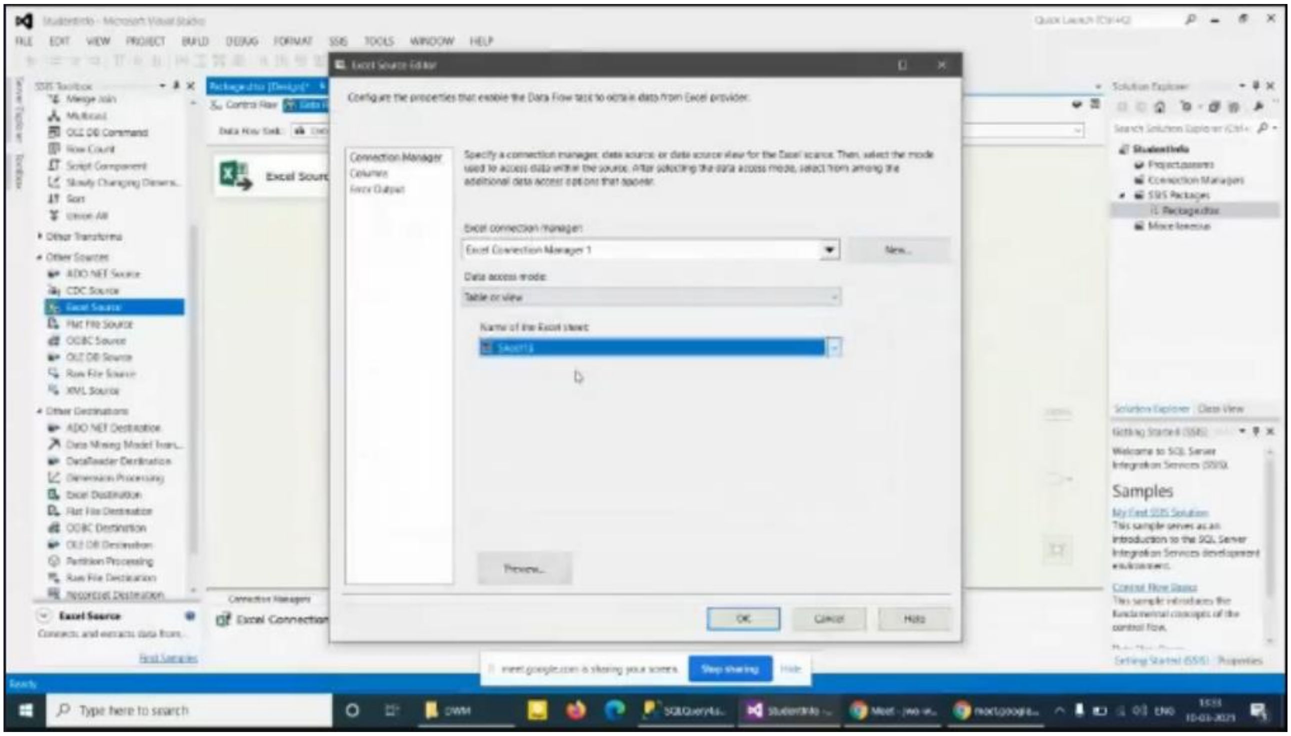
Dataset we will be working with

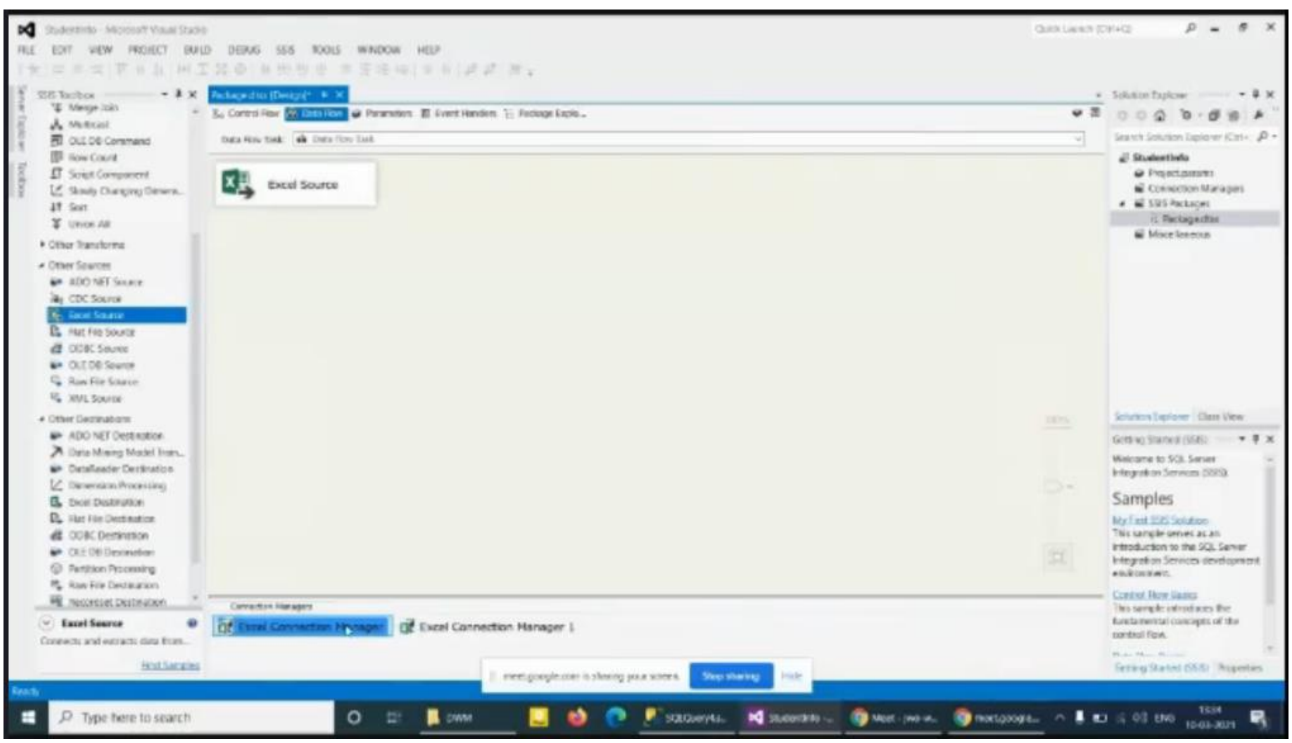


Database in sql

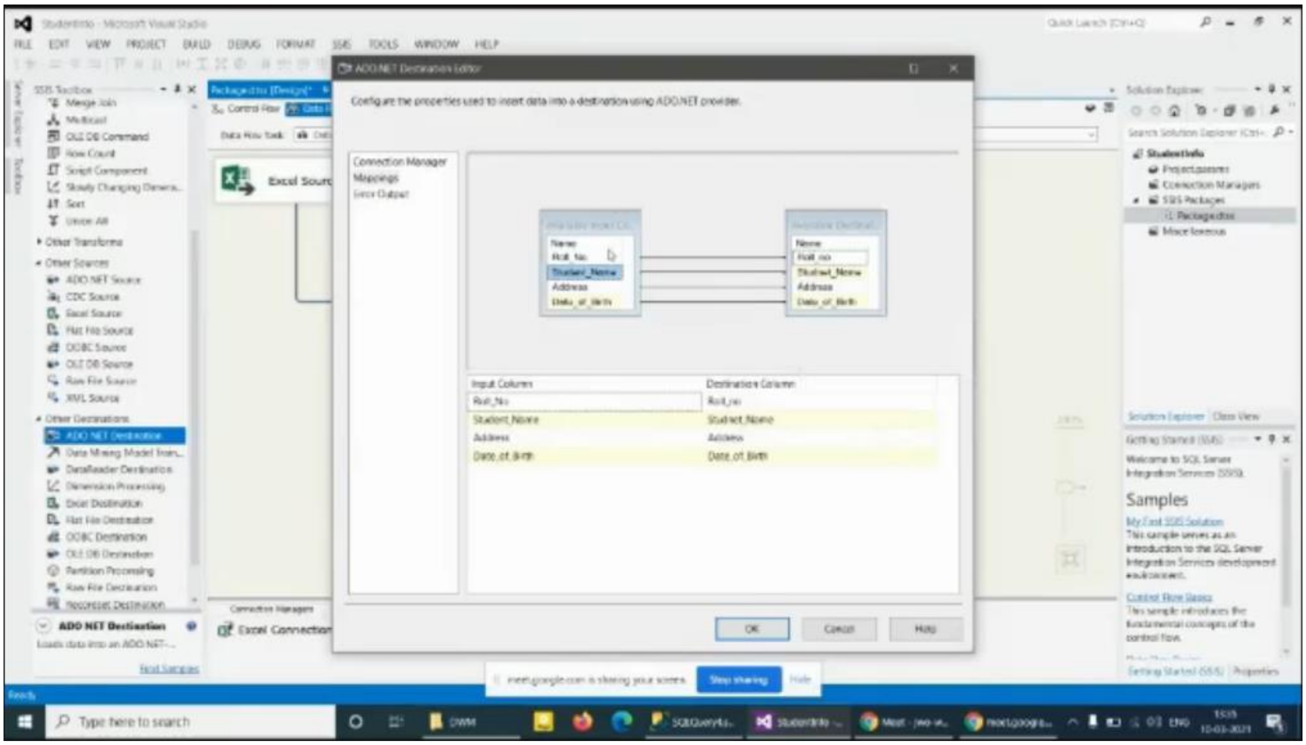


Connecting excel file

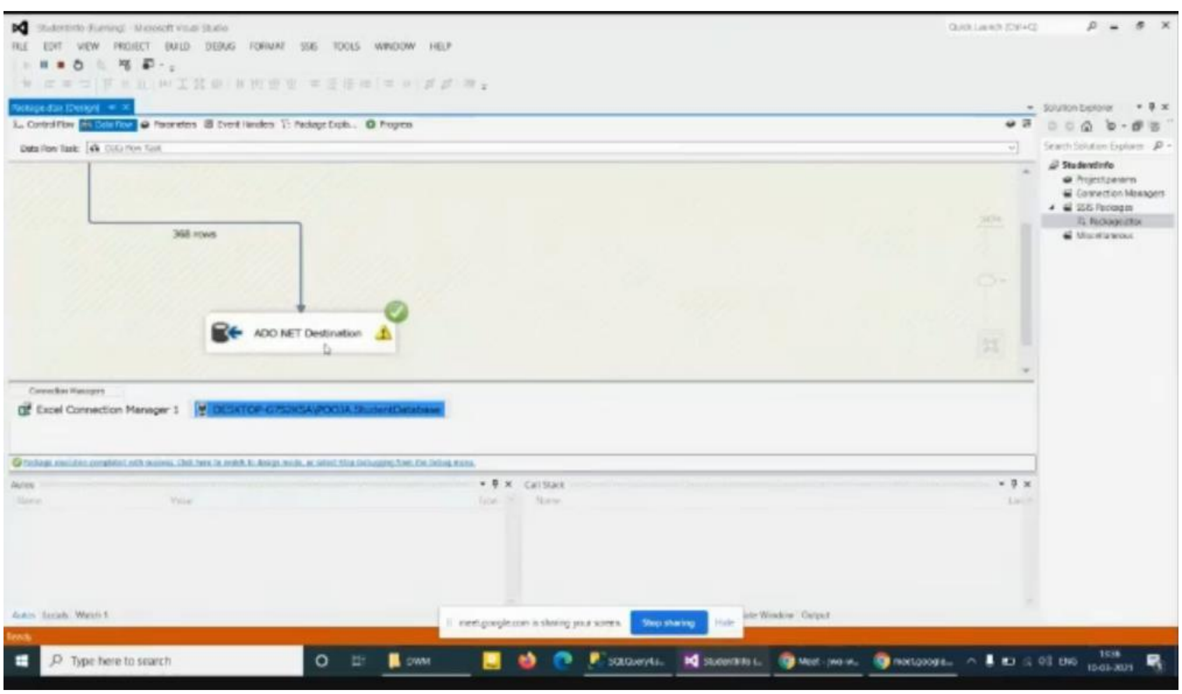




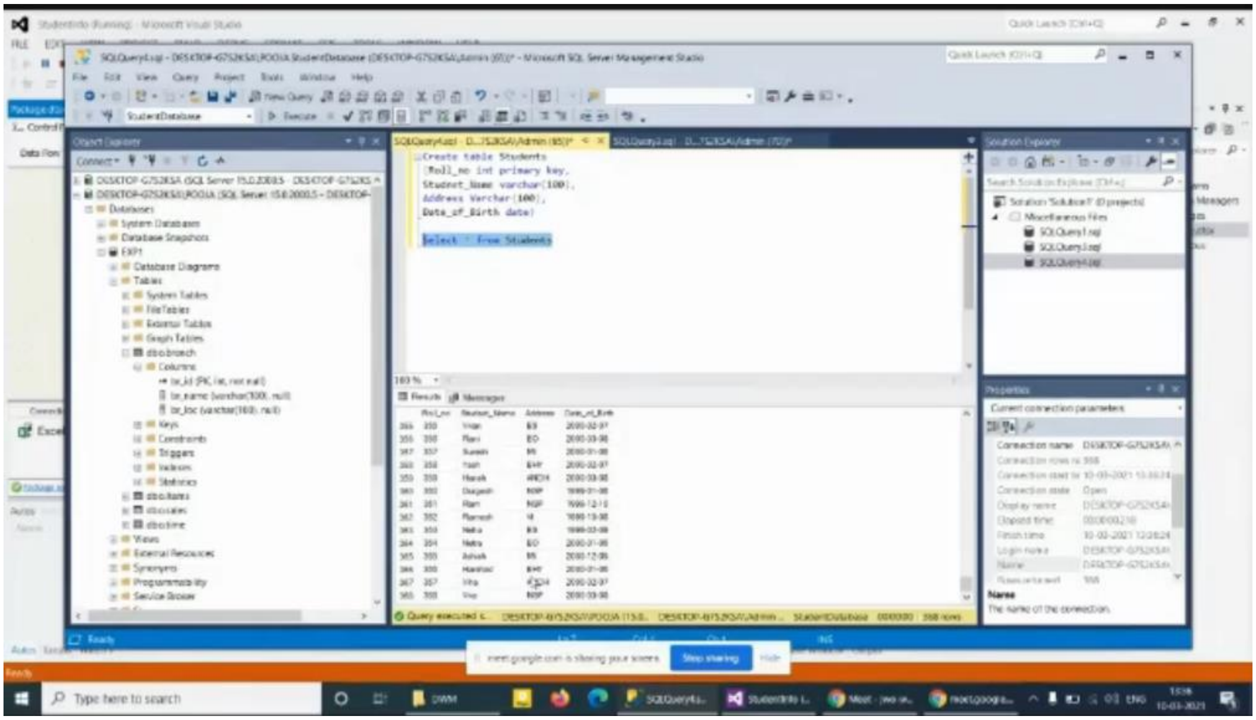
Mapping data



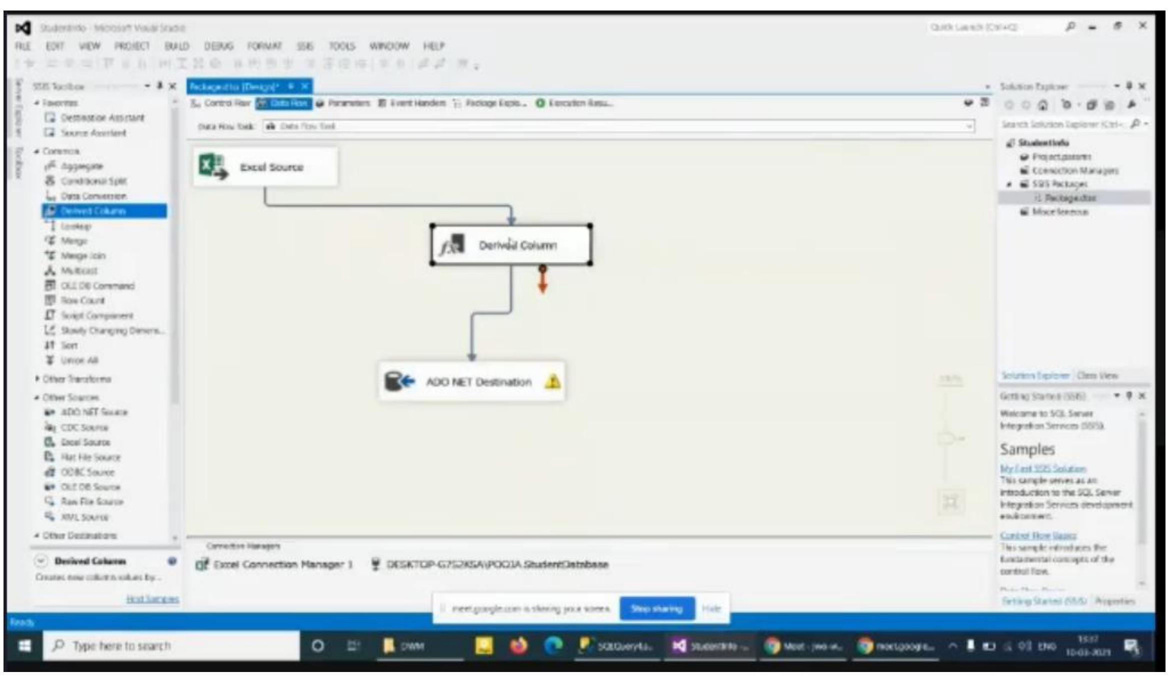
Debugging result



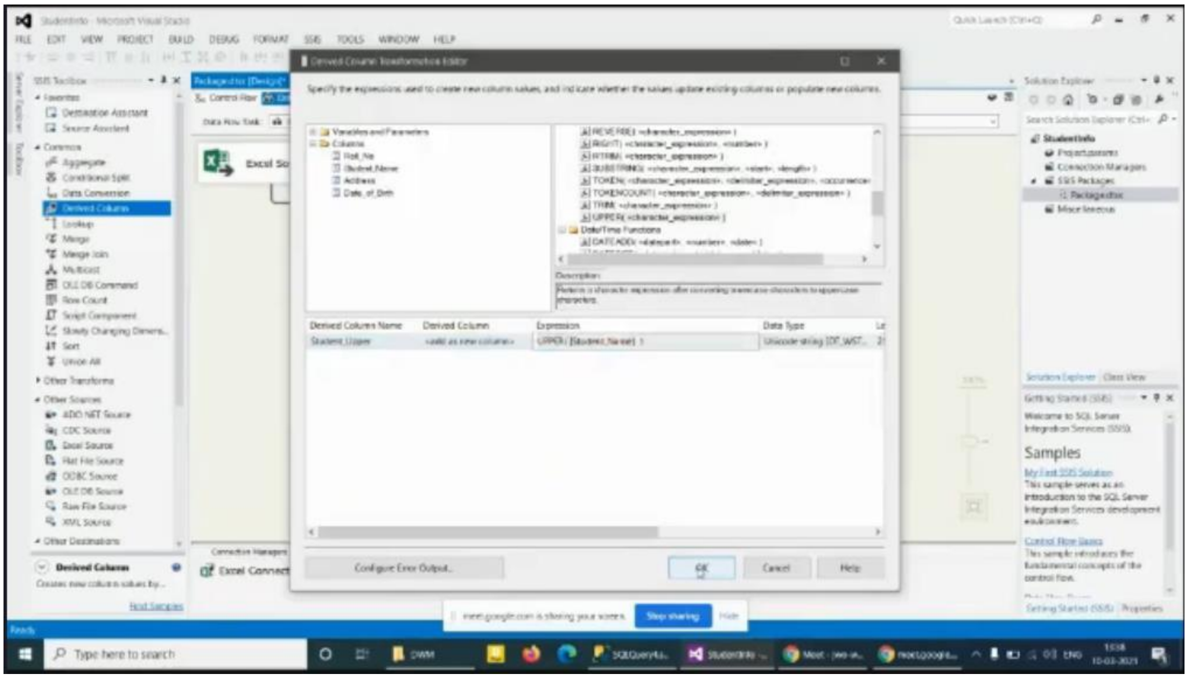
Successfully Loaded Data



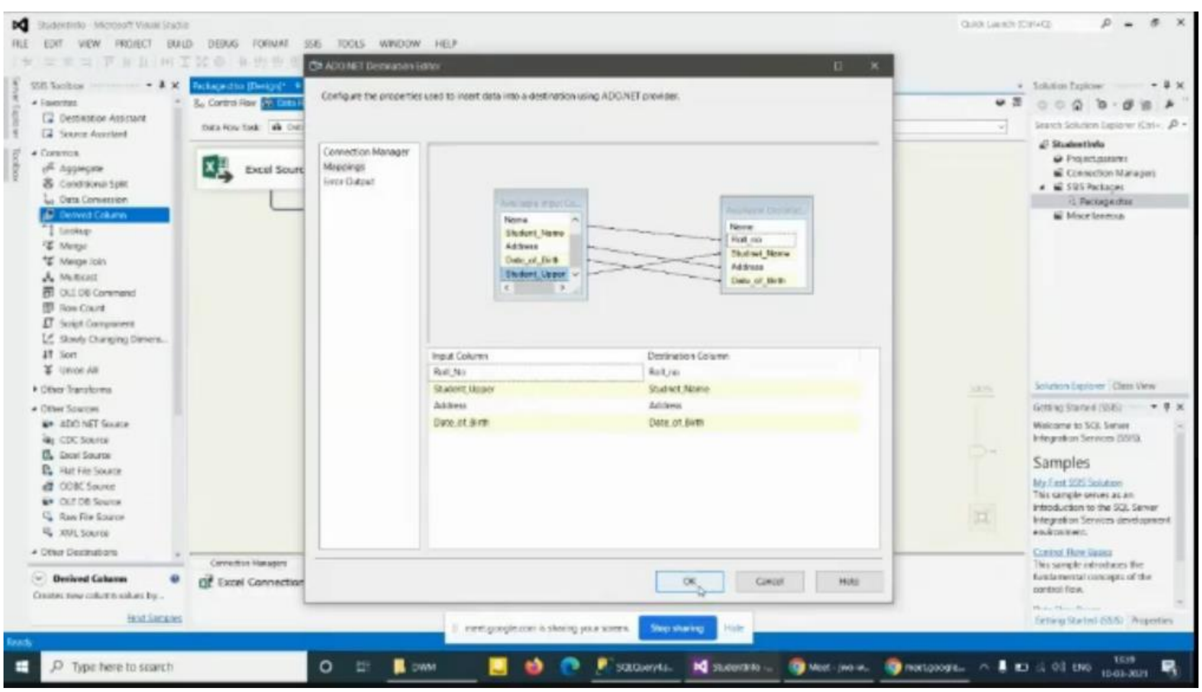
Creating a Derived Column for transformation



Changing Student Name to Upper Case



Change mapping by mapping student upper to student name



After Execution, transformation completed